Bringing Our Expertise to Your Ideas

Agentic AI and SLM Optimization: Designing an AI-Native Architecture for Document Analysis in the Insurance Sector

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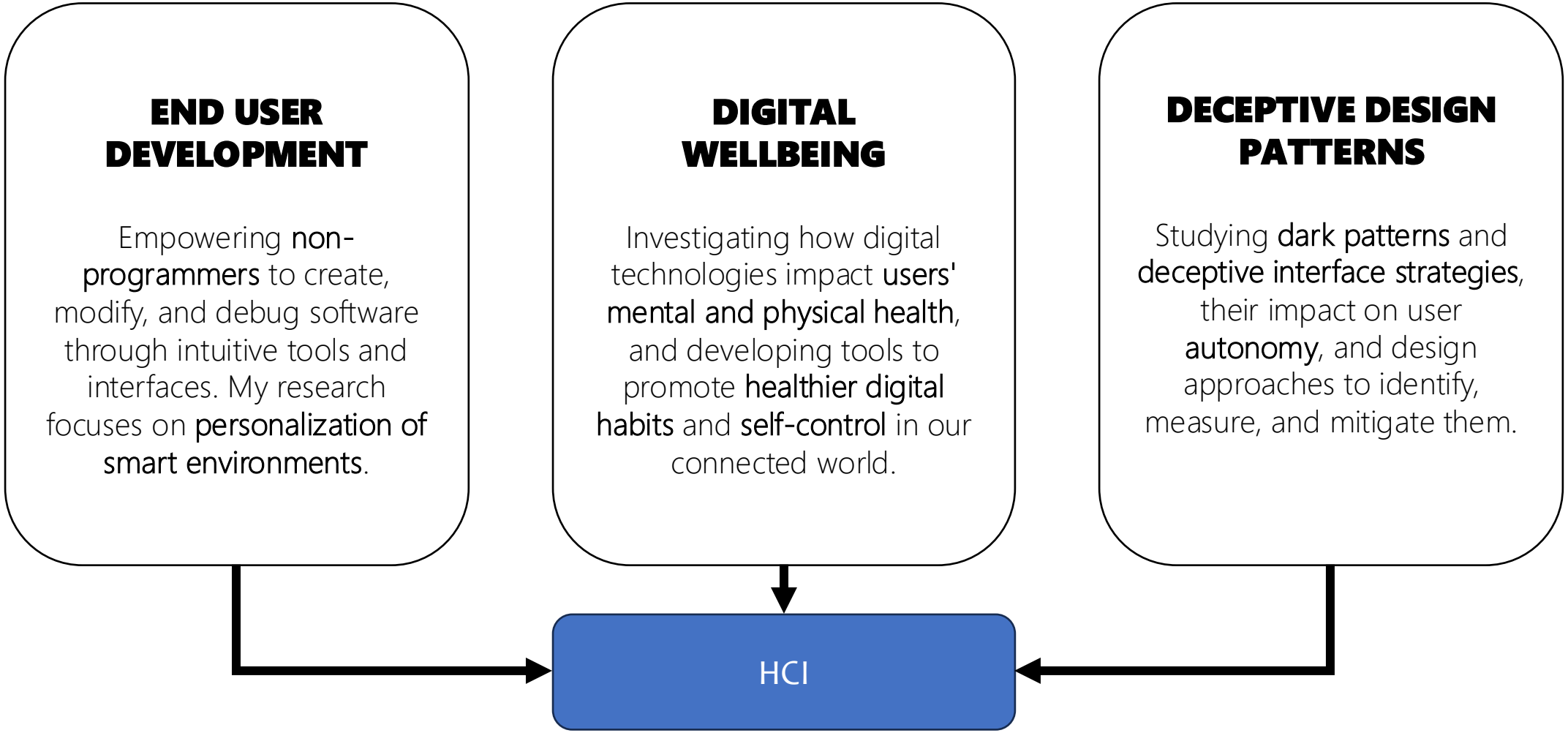
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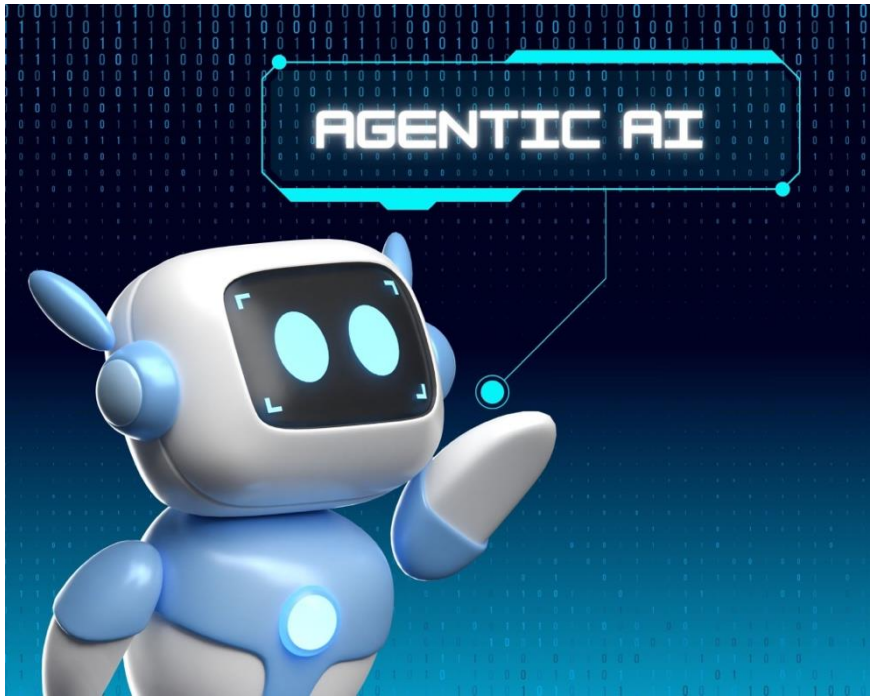
My Research


END USER DEVELOPMENT
Empowering non-programmers to create, modify, and debug software through intuitive tools and interfaces. My research focuses on personalization of smart environments.
DIGITAL WELLBEING
Investigating how digital technologies impact users' mental and physical health, and developing tools to promote healthier digital habits and self-control in our connected world.
DECEPTIVE DESIGN PATTERNS
Studying dark patterns and deceptive interface strategies, their impact on user autonomy, and design approaches to identify, measure, and mitigate them.
HCI

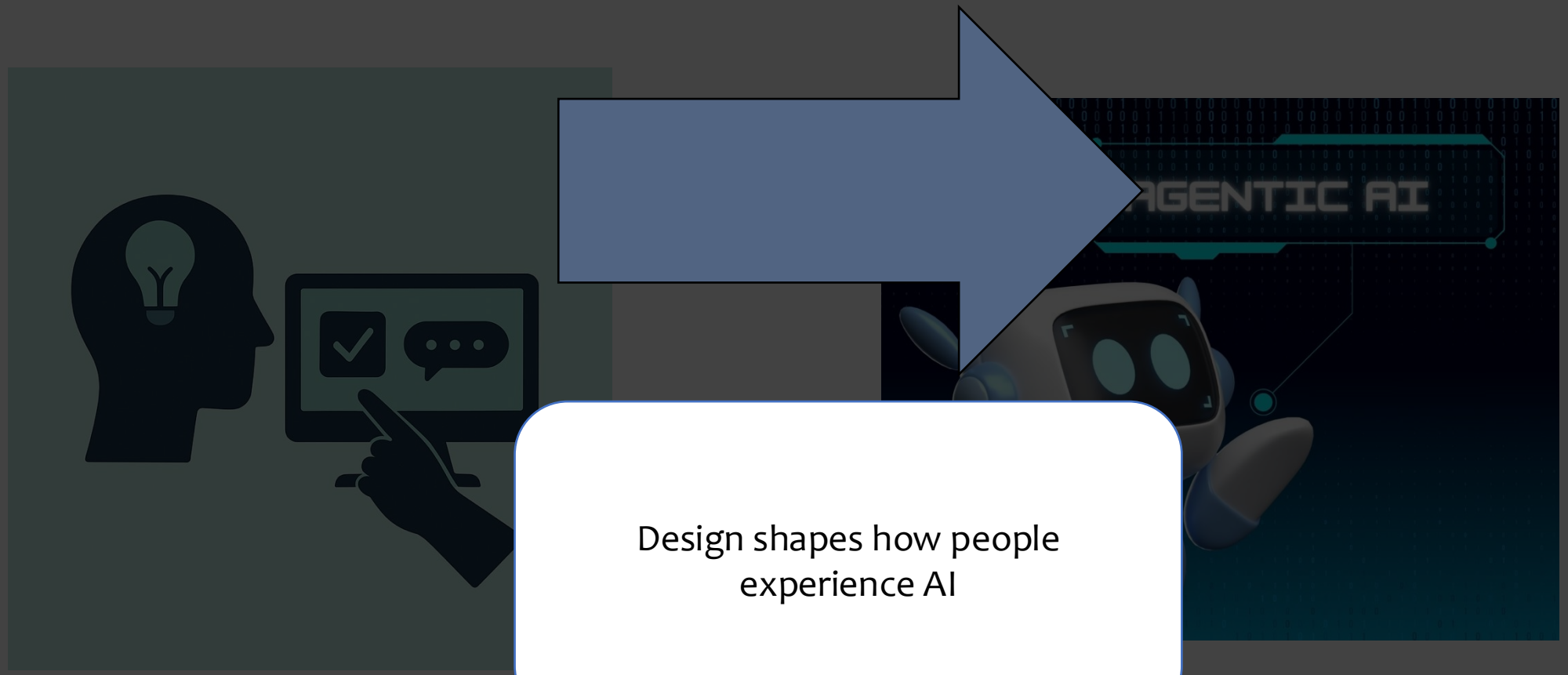
Why HCI Matters for Agentic AI



HCI



Why HCI Matters for Agentic AI

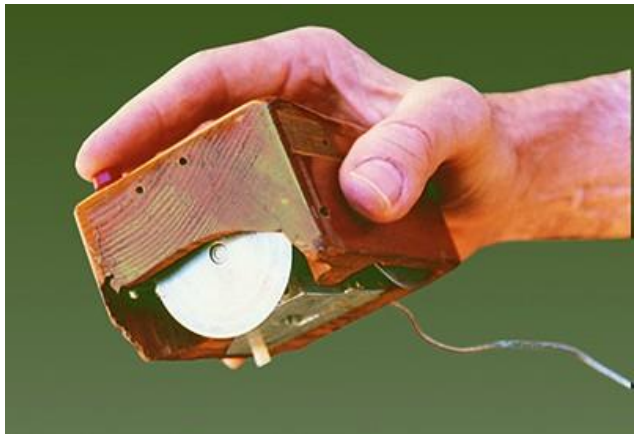


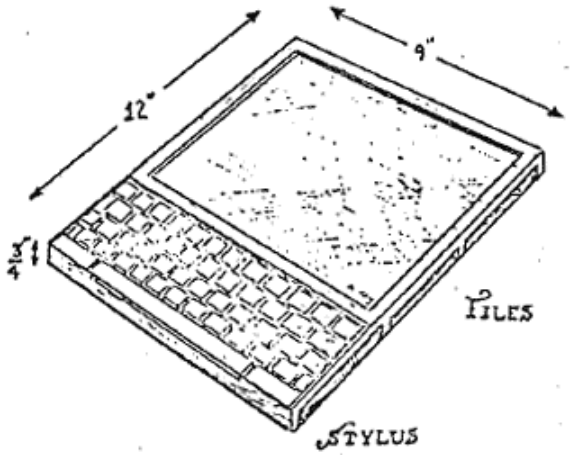
HCI

Design shapes how people experience AI

Human-Computer Interaction… In Brief

- A multi-disciplinary field
- Concerned with the **design**, **evaluation**, and **implementation** of interactive computing systems for human use
  - and with the study of **major phenomena** surrounding them
- Involves two *entities* (the human and the computer) that determine each other behavior over time
  - framed in terms of humans' **goals** and related **tasks/pursuits**





The Goal of HCI

Ingredients

- The **User**(s)
- The **Computer**(s)
- The **Task**(s) to be accomplished

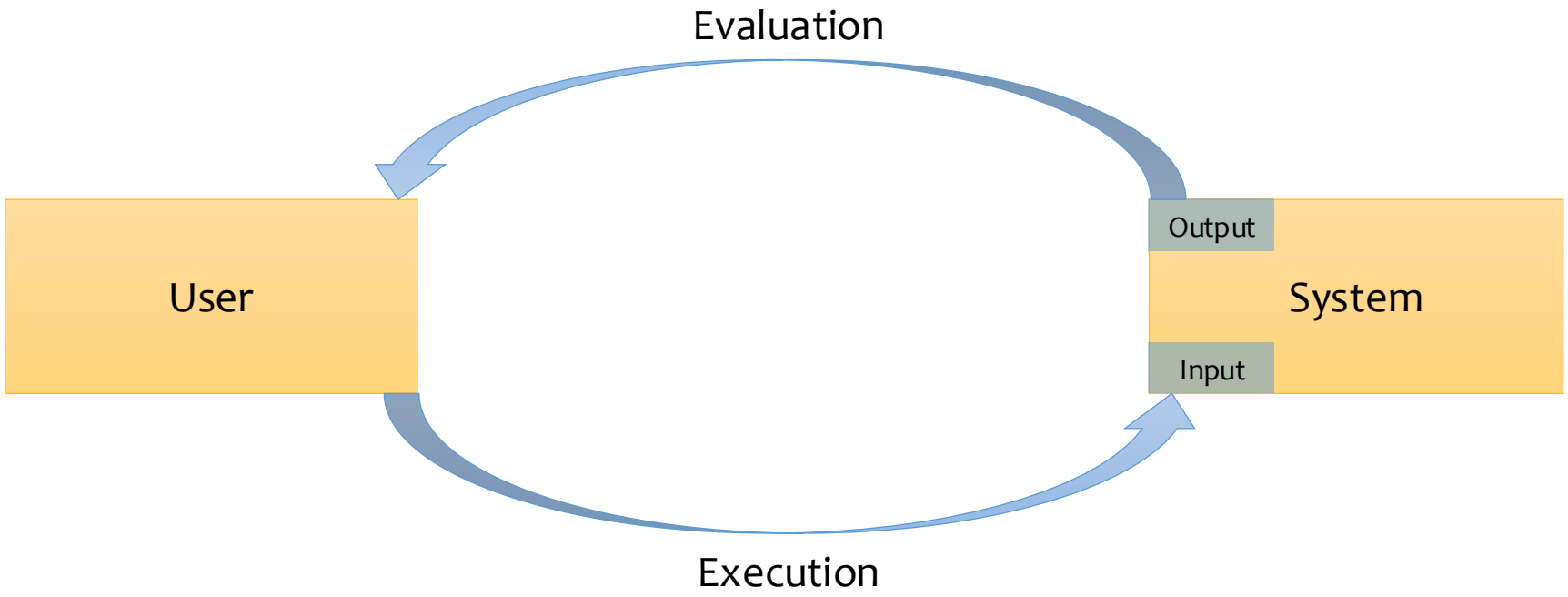
Goal

- The system must support the user's **task**, with a focus on its **usability**
  - Useful
  - Usable
  - Used

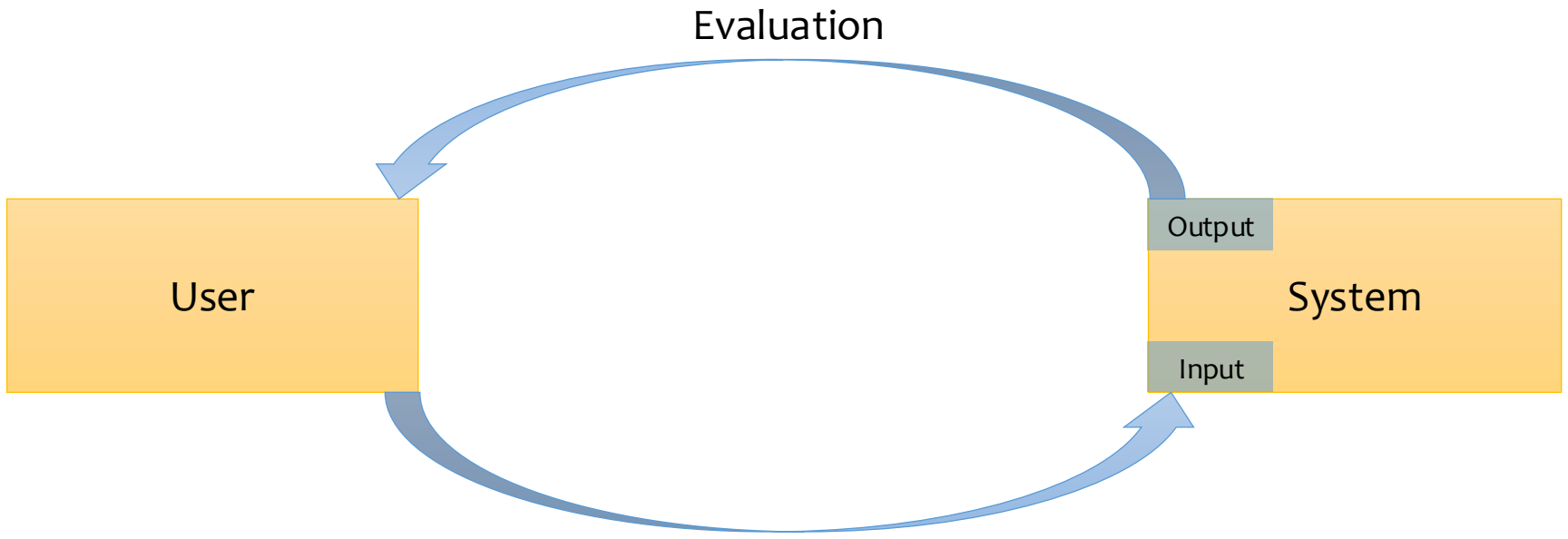
Assumptions

- The **user** wants to accomplish some **goals**, in a specific application **domain**
  - Each domain has a specific jargon, set of possible processes and goals, artifacts and building blocks, …
- **Tasks** are operations to manipulate the concepts of a domain
  - The goal is attained by performing one or more tasks
- Interaction studies the relation between User and System
  - The system possesses a **state** and "speaks" a **core language**
  - The user possesses a **state**, that includes an **understanding** of the system's state, some **intention** to perform a task, and "speaks" the **task language**

Norman's Model of Interaction

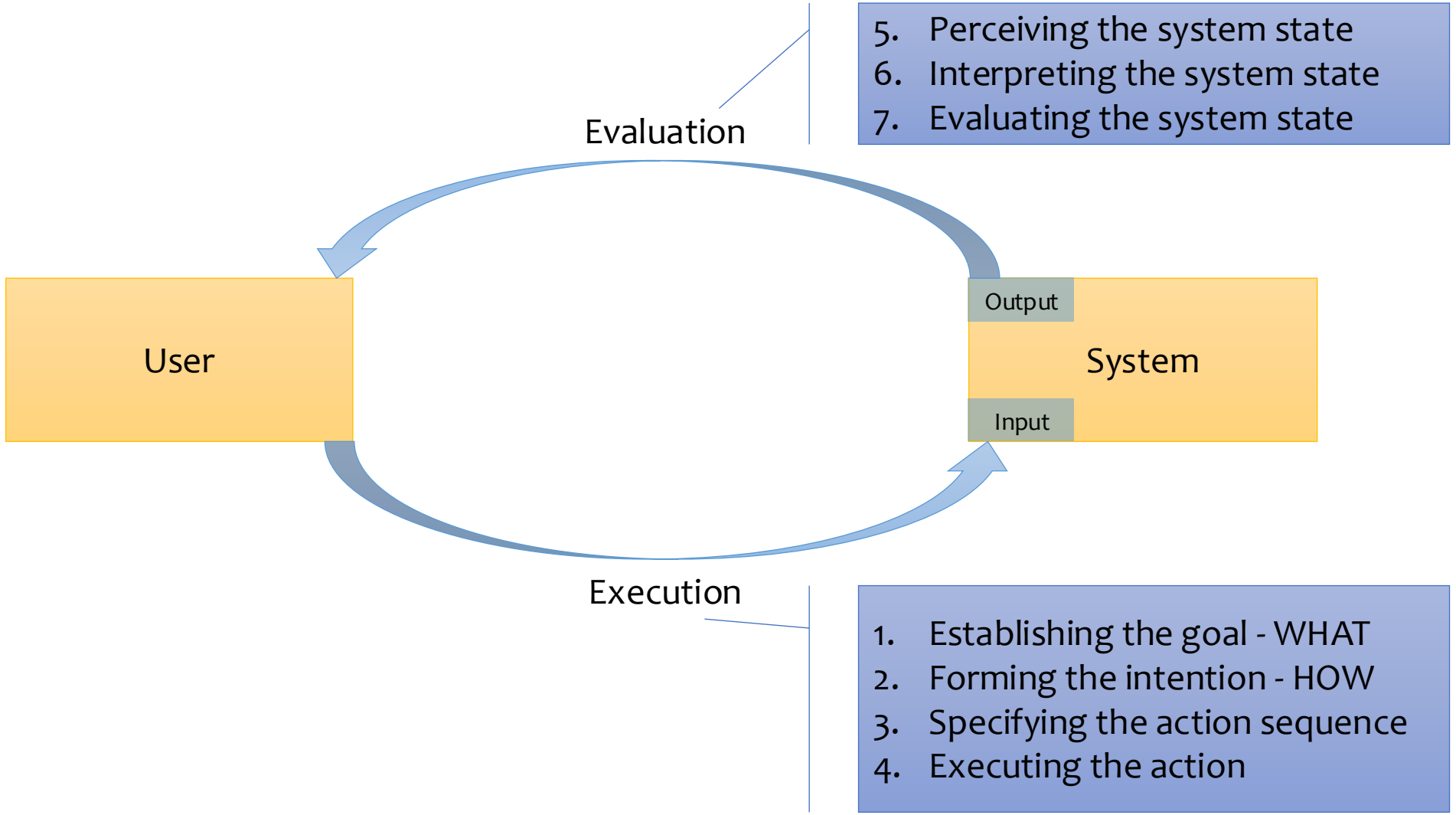


Norman's Model of Interaction

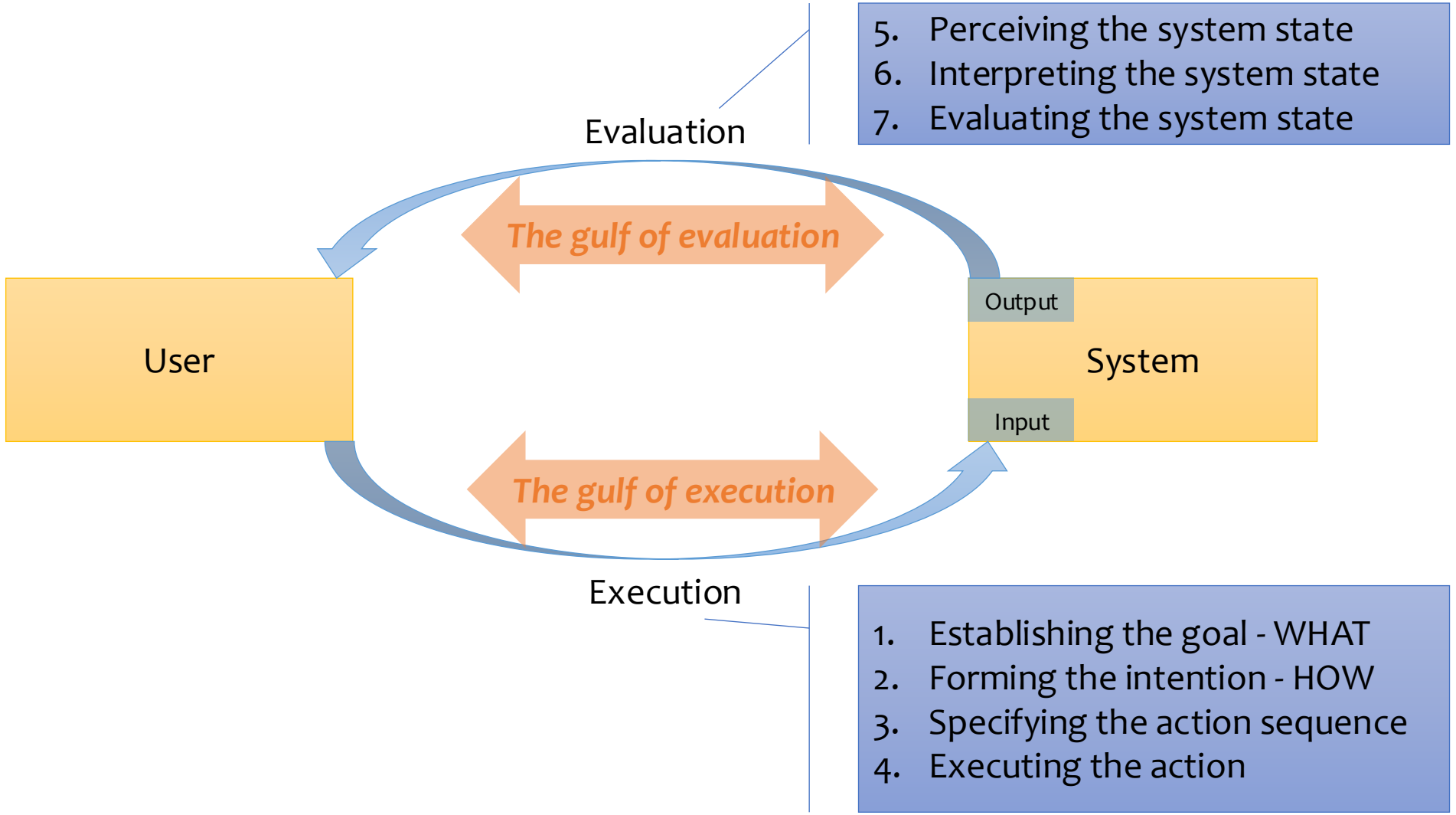


1. Establishing the goal - WHAT
2. Forming the intention - HOW
3. Specifying the action sequence
4. Executing the action

Norman's Model of Interaction



Norman's Model of Interaction



Human Errors*
in the gulf of execution

Slip

- You have formulated the right action, but fail to execute that action correctly
  - E.g., click the wrong icon, or double-click too slow, …
- May be corrected by a better interface (spacing, layout, highlights, …)

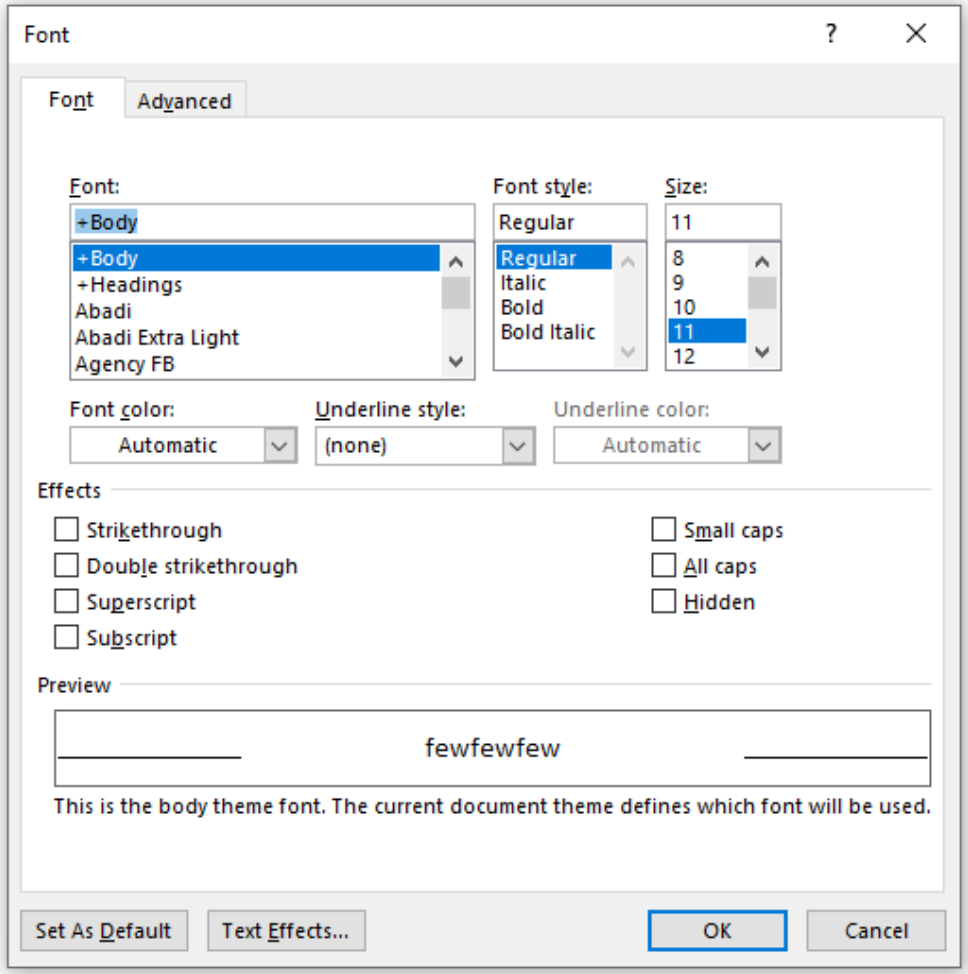
Mistake

- You don't know the system well and you may not formulate the right goal
  - E.g., click 🔎 for Zoom, but it means Search
- The user's mental model of the system's state is not correct
- Requires more radical redesign, or additional training

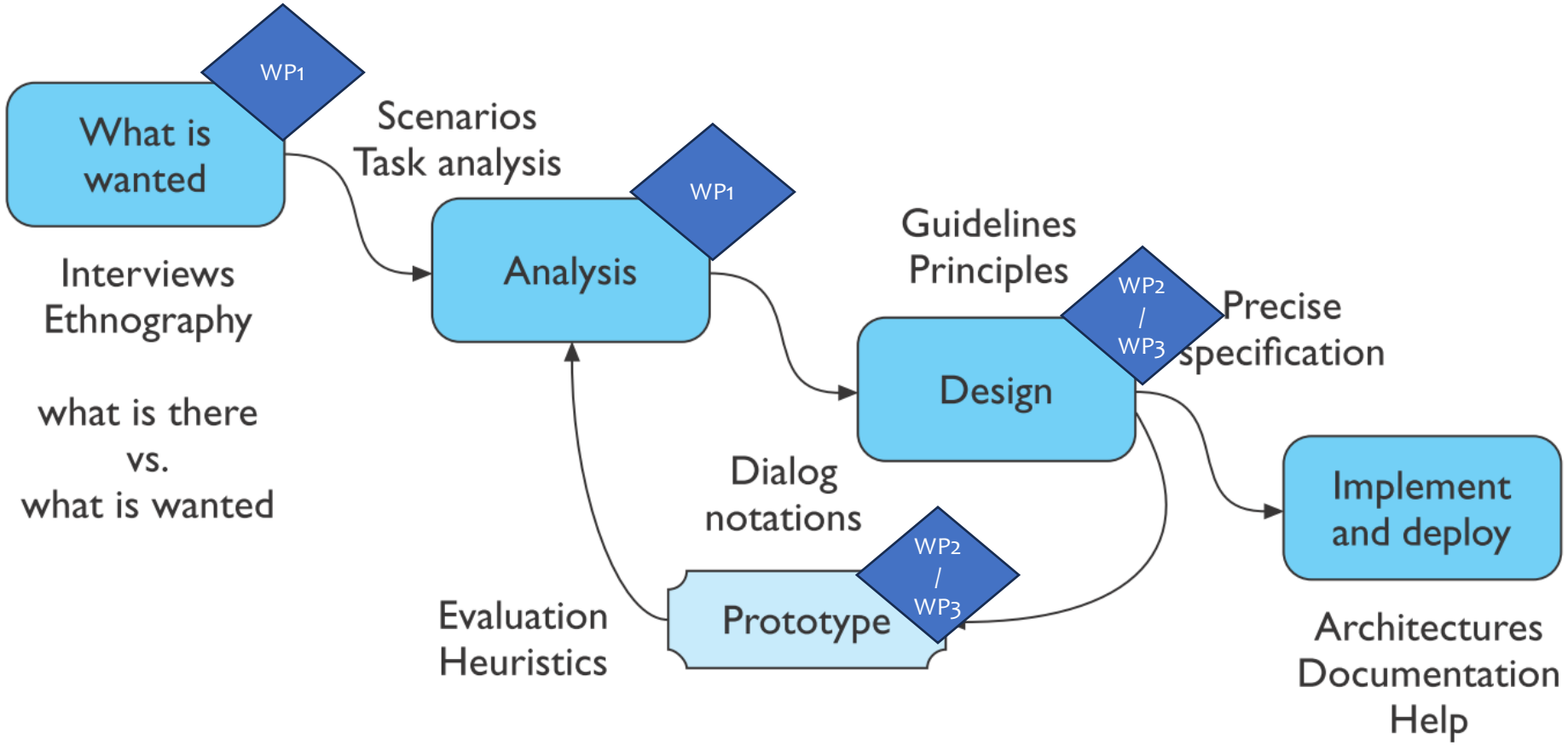
* About Human Errors

- Human errors should **never** be considered as faults of the user
- Rather, «they are usually a result of bad design» (Norman)
- Humans tend to be imprecise, distracted, not-omniscient
  - System design should anticipate this human behavior
  - Minimize the chance of inappropriate actions (evaluation)
  - Maximize the possibility of discovering and repairing an inappropriate action (execution)
  - Enable users to understand the state of the system and build an appropriate model

Example:
Which are the allowed combinations?

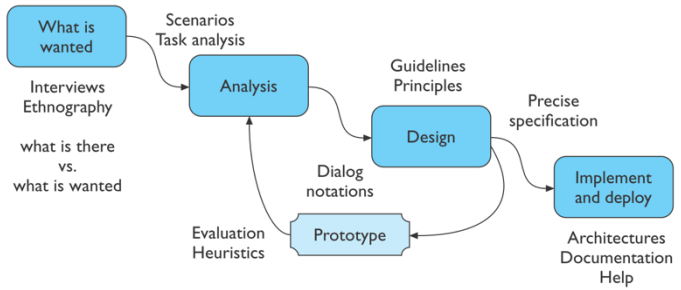


Human-Centered Design Process
(simplified and generic)



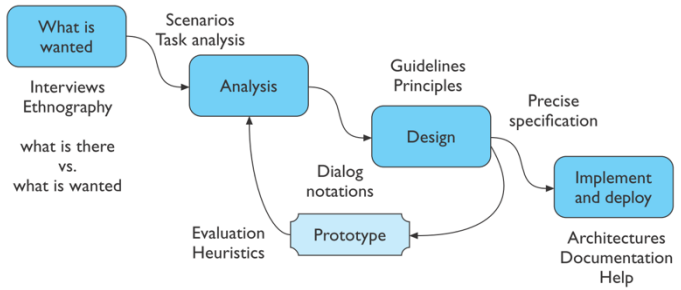
Human-Centered Design Process – The Main Steps (1)

- Needfinding – what is needed by people
  - What exactly is needed? How are people currently accomplishing the goal?
  - User observation, interviews, …
- Analysis
  - Formalize and structure the needs
  - Create interaction scenarios, stories, tasks
  - Compare current situation with expected new situation

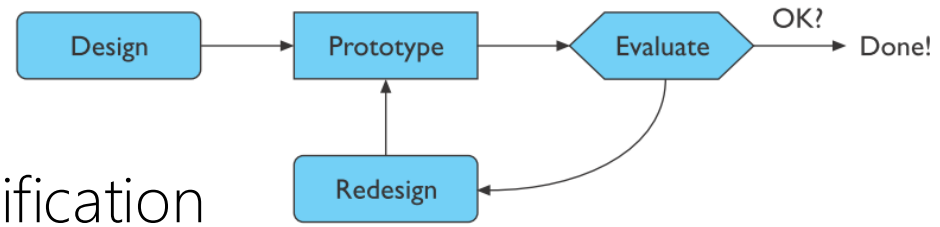


Human-Centered Design Process – The Main Steps (2)

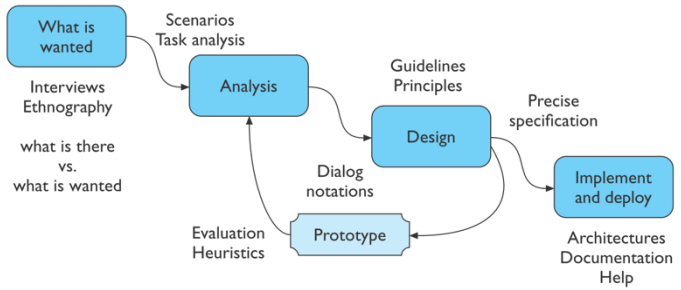
- Design
  - The main choices to shape the system
  - Rules, guidelines, design principles
  - Considering different types of users
  - Modeling and describing interaction
  - Visual layout
  - Consider all inputs from cognitive models, communications theories, organization issues



Human-Centered Design Process – The Main Steps (3)



- Iteration and prototyping
  - Design must be supported by intermediate verification
  - Evaluate the design in its partial forms:
    - Prototypes
    - Evaluation metrics
  - Involving users
- Implementation and deployment
  - Hardware and software implementation
  - Documentation



Human-AI Interaction vs Human-Computer Interaction?

- What is different when AI is in an interactive system?
- AI-based systems are typically performed under **uncertainty**
  - often producing false positives and false negatives
- Many AI components are inherently **inconsistent**
  - they may respond differently to the same text input over time (e.g., autocompletion systems suggesting different words after language model updates)
  - or behave differently from one user to the next (e.g., search engines returning different results due to personalization)

Can AI and People Really Work Together?

1. Should we design and develop technology that
   - automatizes people's actions, replaces humans (*Artificial Intelligence*)
   - or augments them (*Intelligent Augmentation*)?

Is there a clear winner?

2. Which kind of human-AI collaboration can we envision?
   - full human control… no human involvement?
   - full AI autonomy… no AI?
   - in which cases?

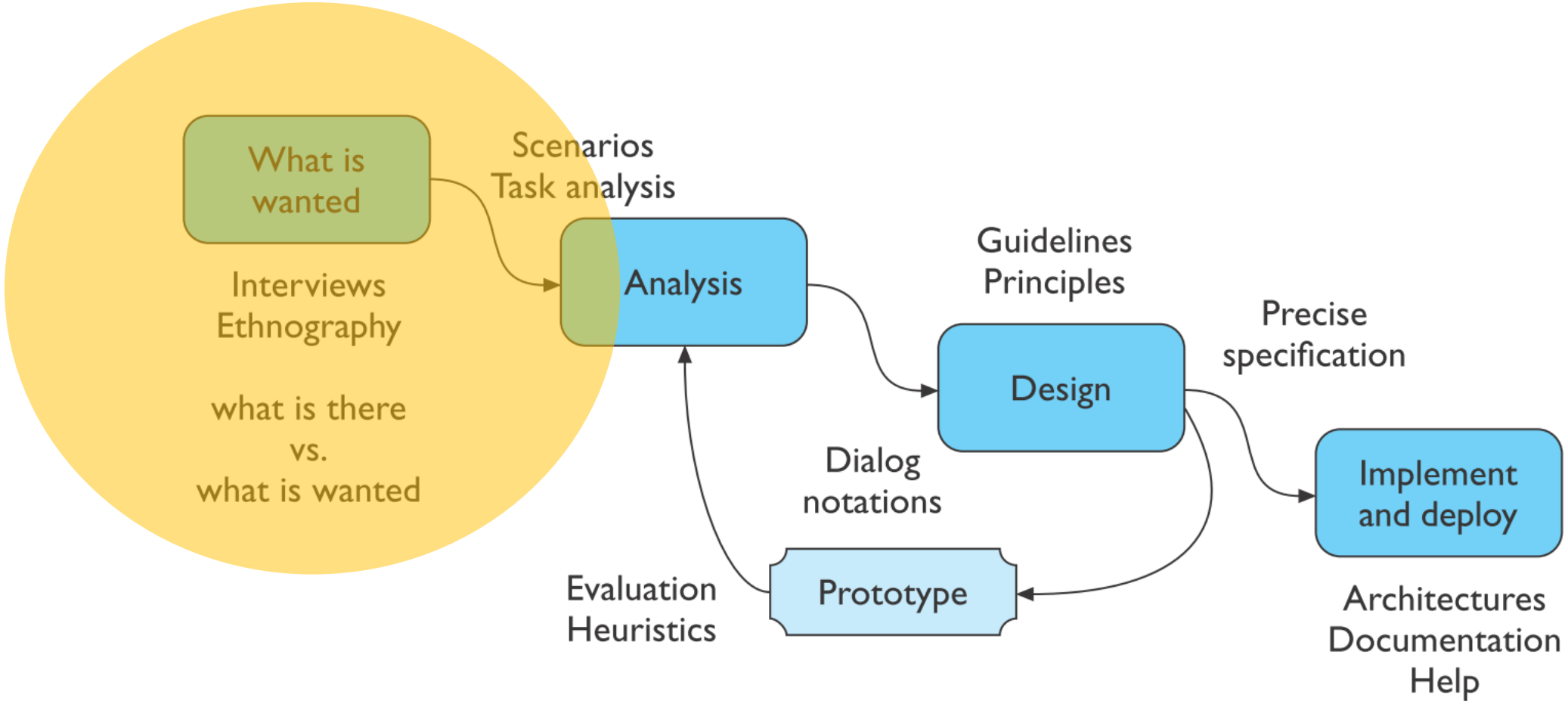
Traditional Guidelines and AI

- AI-based systems can also violate established usability guidelines of traditional user interface design
  - for instance: consistency or error prevention
- Many AI components are inherently **inconsistent**
  - they may respond differently to the same text input over time (e.g., autocompletion systems suggesting different words after language model updates)
  - or behave differently from one user to the next (e.g., search engines returning different results due to personalization)

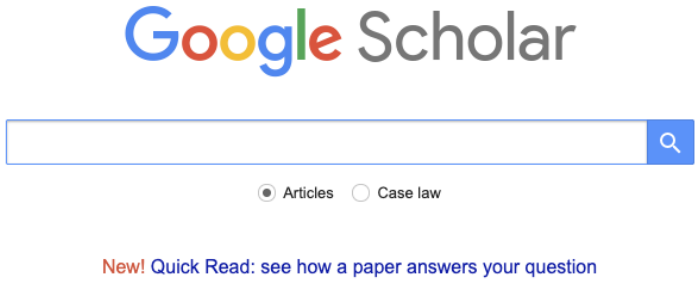
How Can We Design Interactivity with AI?

- In brief, case by case, feature by feature:
  - Decide when "to AI" and when "not to AI"
  - Understand when to automate (i.e., replace the user) and when to augment users' capabilities
  - Balance the uncertainty of AI systems with proper expectations and feedback
- In general, the more **critical** a functionality is, the more people *need* accurate and reliable results
- On the other hand, if a **complementary** feature delivers results that are not always of the highest quality, people *may* be more forgiving

Needfinding



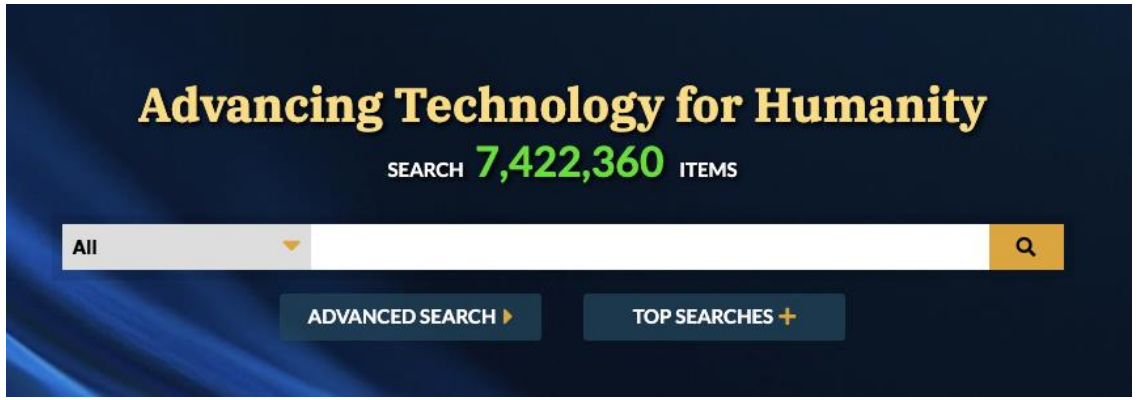
Bibliographic Research



https://scholar.google.com/



https://dl.acm.org/



https://ieeexplore.ieee.org/

Main Forms of Interviews

- In-person interviews
  - Time-demanding, in-depth knowledge
  - Structured vs Unstructured
  - One-to-one vs. Focus groups
- Surveys
  - Fast, more superficial
  - Sets of questions with predefined possible answers
  - Paper-based or on-line

Beware!

- Users don't know what they want
  - Maybe subconsciously, but not rationally
  - They will tell what they think you like to hear
  - Especially for "new" products or "disruptive" technologies
  - They lack the creativity or the technical expertise to understand the new product
  - They take the current context for granted (e.g., required workarounds "because, yes")



Stop Asking Users What They Want
https://uxplanet.org/stop-asking-users-what-they-want-21e9ba646bce

Guidelines For Questions in Interviews

- Structured questions are easier to process, unstructured questions solicit more comments
- Open-ended questions, with follow-up discussion
- For quantitative questions (e.g., rate in a scale 1 to 5, such as Likert), always ask what they mean by "4"
- Aim at direct, concrete, specific questions that ask for detailed answers
- Use the language of the user
- Always try the question with a smaller (trusted) group, for debugging

Bad questions in interviews – To Avoid

- Why is feature [x] important to you?
  - ‘Leading’ question
- What would you like in a tool?
  - User are experts in their domain, not expert in design
- What do you like in [x]?
  - Assuming question. Maybe he doesn’t like it
- What would you do in a hypothetical situation?
  - Users cannot imagine the complete environment or an unusual situation

Bad questions in interviews – To Avoid

- How often do you do [x]?
  - Humans are very bad at estimating (and biased in the answers)
  - May obtain by log analysis (if an application already exists)
- Binary questions (yes/no)
  - Do not yield motivation
- 'Tell me a story about you'
- 'How do you reach the decision? Did you meet? Did someone decide without you? …'
  - Avoid suggesting possible answers. Trust the question: ask it and wait in silence

On-line surveys

- Familiar, cost-effective, potentially reaching a very wide audience
- Results can be easily visualized and analyzed with statistical methods

- Should be very careful with setting the goal of the survey
  - First define what statistics and/or charts you need, then design the survey structure and questions

Risks in surveys

- Good for a shallow view over a large base of respondent, but not for a "deep" analysis
- Impossible to ask follow-up questions
- Biased data if questions rely on user's memory or on "sensitive" issues (money, emotions, …)
- Finding a representative target population
  - Stratification

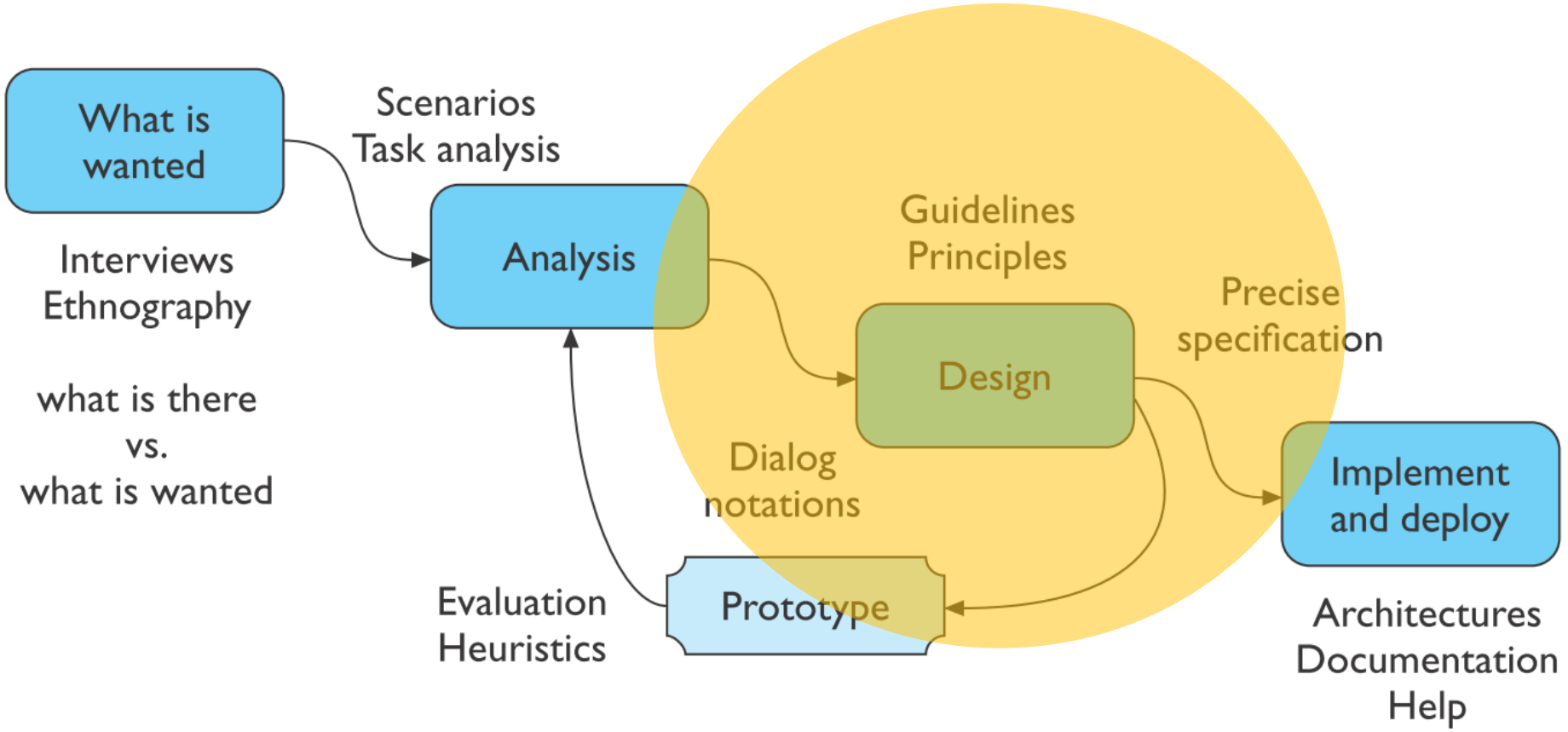
Survey structure

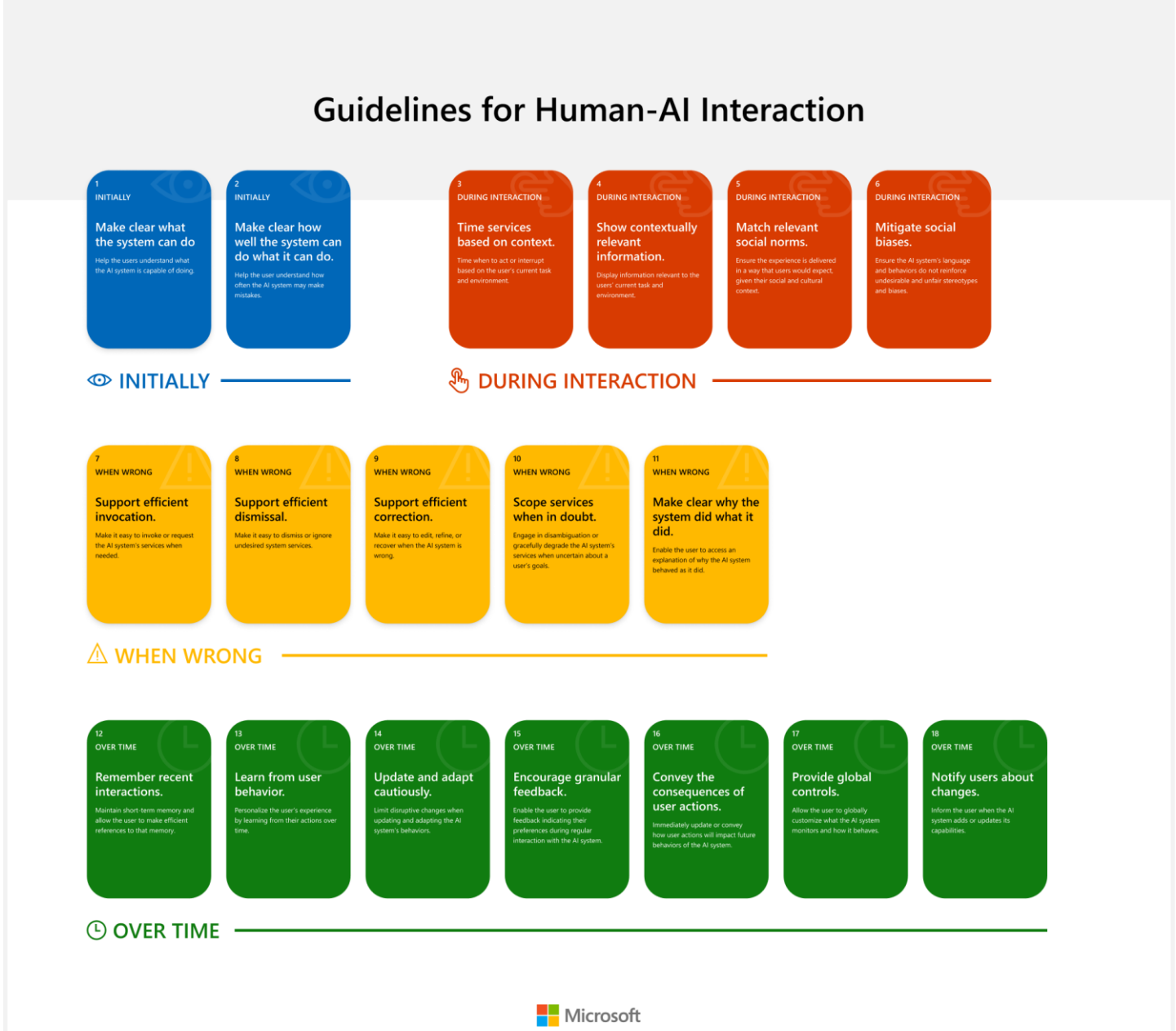
- Declare the purpose of the survey and the expected time
- One or more sections with targeted questions
- Background information about the users
  - Limit “mandatory” fields to a minimum

Types of questions for surveys

- Open-ended questions
  - Solicit specific motivation, to avoid too generic answers
  - Require methodology for analyzing the responses
- Closed-ended questions
  - Only one possible choice
  - **Ordinal** values: a scale of ordered possibilities (e.g., from 1 to 5)
  - **Nominal** values: alternatives, with no ordering relationships (e.g., city or department)

Design for Human-AI Interaction





By Microsoft Research: https://www.microsoft.com/en-us/research/project/guidelines-for-human-ai-interaction/

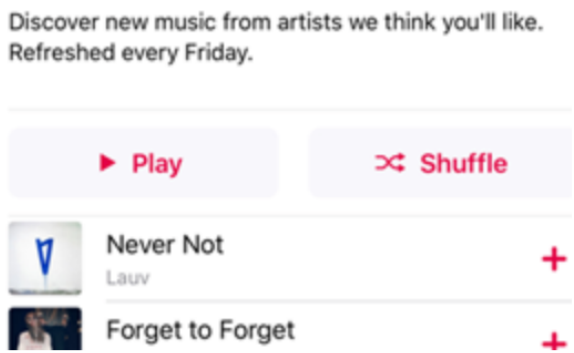
2

INITIALLY

Make clear how well the system can do what it can do.

Help the user understand how often the AI system may make mistakes.

EXAMPLE IN PRACTICE



The recommender in **Apple Music** uses language such as "we think you'll like" to communicate uncertainty.

Make clear how well the system can do what it can do. 2

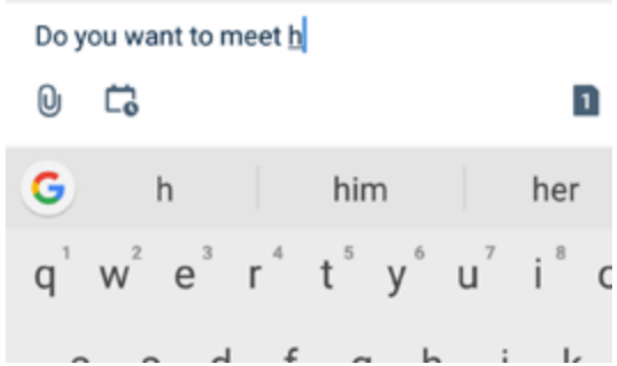
6

DURING INTERACTION

Mitigate social biases.

Ensure the AI system's language and behaviors do not reinforce undesirable and unfair stereotypes and biases.

EXAMPLE IN PRACTICE



The predictive keyboard for **Android** suggests both genders when typing a pronoun starting with the letter "h."

Mitigate social biases. 6

9

WHEN WRONG

Support efficient correction.

Make it easy to edit, refine, or recover when the AI system is wrong.

EXAMPLE IN PRACTICE



When **Bing** automatically corrects spelling errors in search queries, it provides the option to revert to the query as originally typed with one click.

Support efficient correction. 9

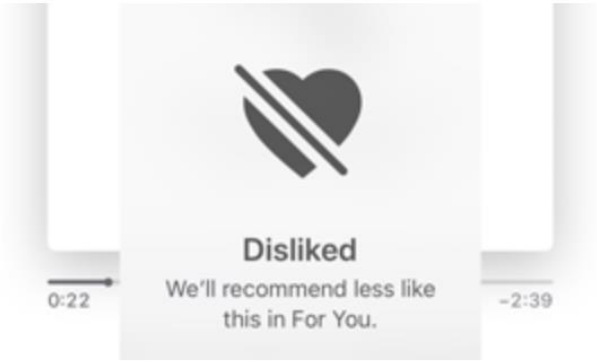
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OVER TIME

Convey the consequences of user actions.

Immediately update or convey how user actions will impact future behaviors of the AI system.

EXAMPLE IN PRACTICE



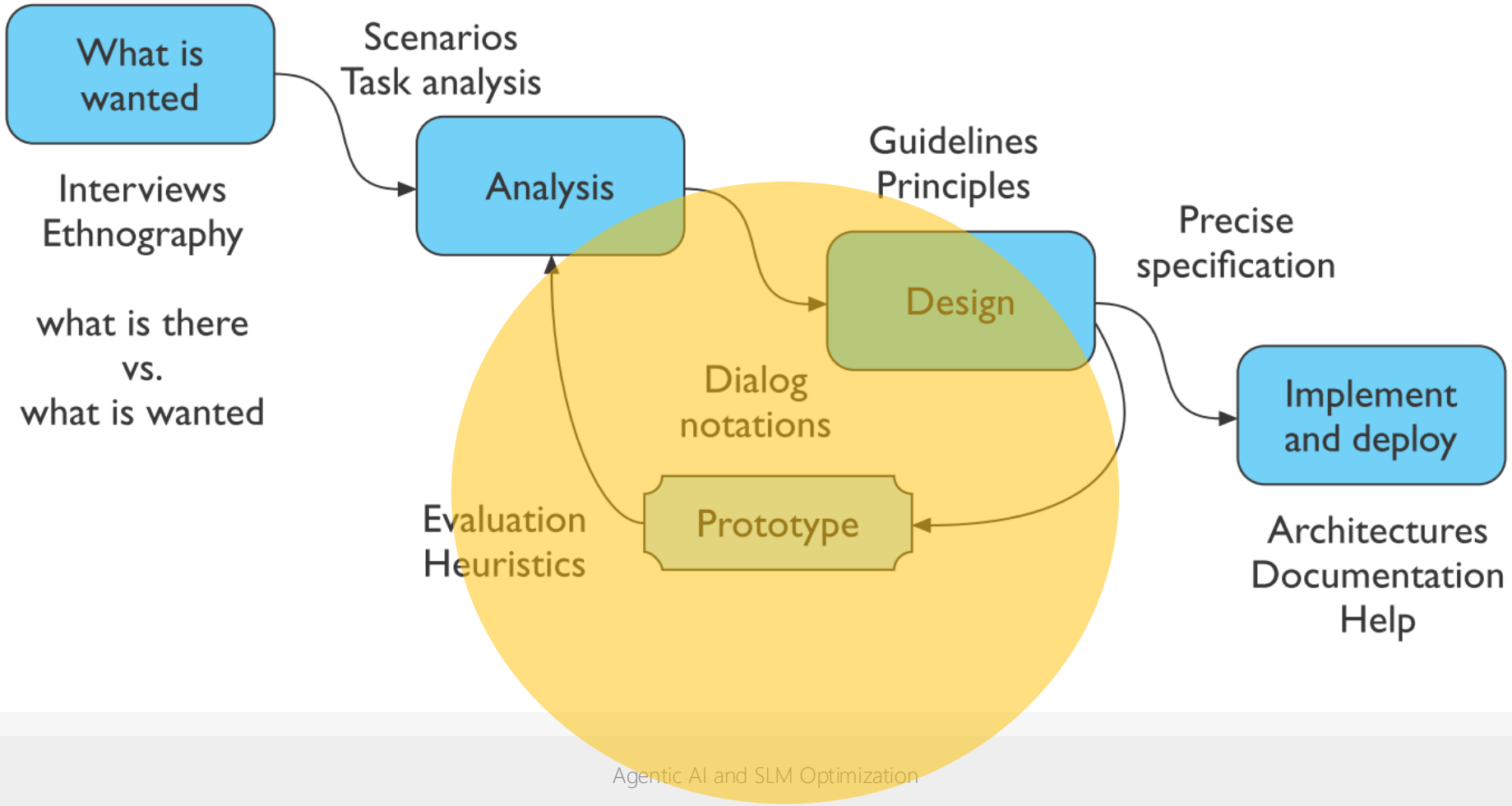
Upon tapping the like/dislike button for each recommendation in **Apple Music**, a pop-up informs the user that they'll receive more/fewer similar recommendations.

Convey the consequences of user actions. 16

Other Guidelines

- Design Principles for Generative AI Applications: https://dl.acm.org/doi/10.1145/3613904.3642466
- Google's People+AI Guidebook: https://pair.withgoogle.com/guidebook/
- Apple's Human Interface Guidelines for Machine Learning: https://developer.apple.com/design/human-interface-guidelines/machine-learning/
- Microsoft's Human-AI eXperience Toolkit: https://www.microsoft.com/en-us/haxtoolkit/

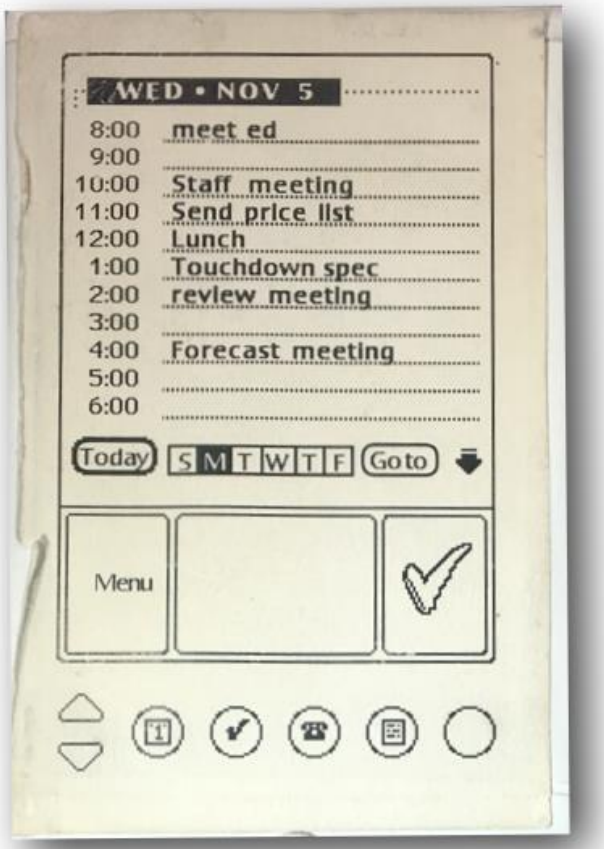
Prototyping Human-AI Systems



Prototypes

- «A prototype is a concrete but partial representation or implementation of a system design»
- «An easily modified and extensible model (representation, simulation or demonstration) of a planned software system, likely including its interface and input/output functionality»
- One of the most powerful tools for design exploration, visualization, and testing
- They let us ‘see’ and ‘feel’ interactivity (simulated or real)

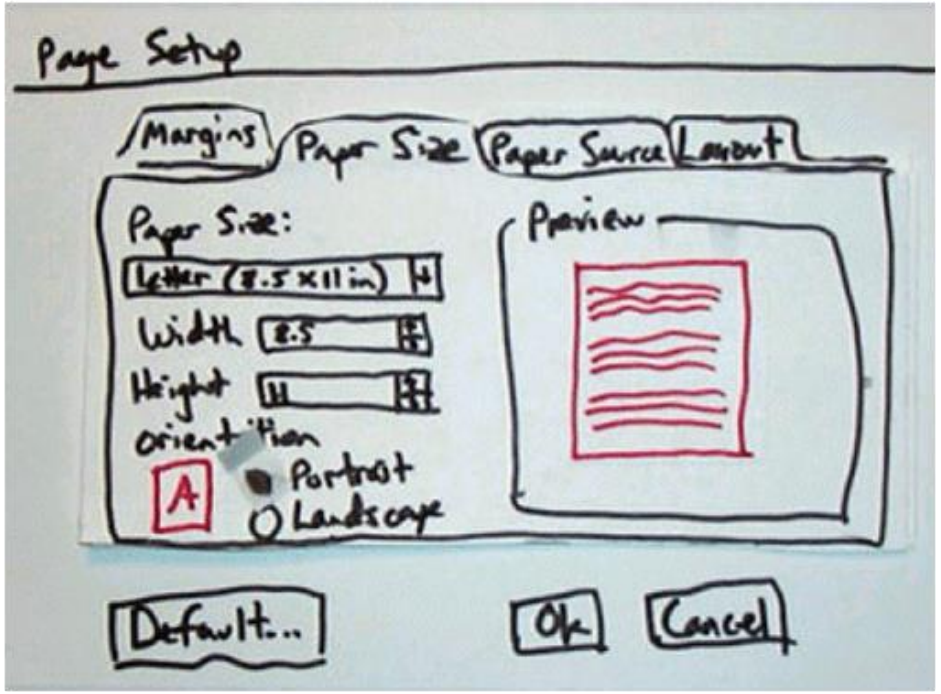


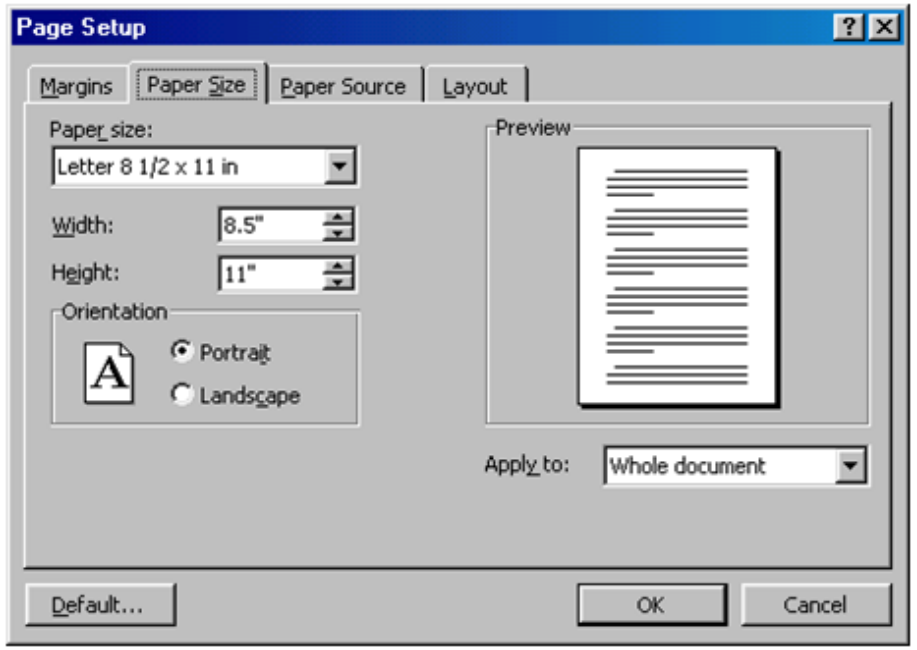




source: https://albertosavoia.medium.com/the-palm-pilot-story-1a3424d2ffe4

Fidelity: Different Information Is Conveyed





Projects Completion Level (I)

- The projects will deliver **prototypes**, meaning that:
  - They will be implemented using any technology you prefer or are familiar with, following the company's recommendations
  - They won't be a final "product"
  - They will simulate a realistic experience

- For example, you can use web technology to built a prototype that *looks like* and *behave as* a mobile application, or you can develop a SLM on a limited subset of data or for a specific use case

Projects Completion Level (II)

- Prototypes may make assumptions and omit standard or highly complex features, even if they are technically important:
  - Which will be *faked*, *limited* in number, or *hard coded*
  - Any particular behavior will work in a small number of pre-defined cases
  - …you can identify these assumptions by discussing them with the company and your mentors!

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Up Next: Meet Your Mentors!

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